

Day 6 Of Communication System Class

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6.4 Line coding schemes:

Digital-to-Digital Encoding

Digital-to-Digital Encoding is the representation of digital information by a digital signal. (eg. computer-to-printer)

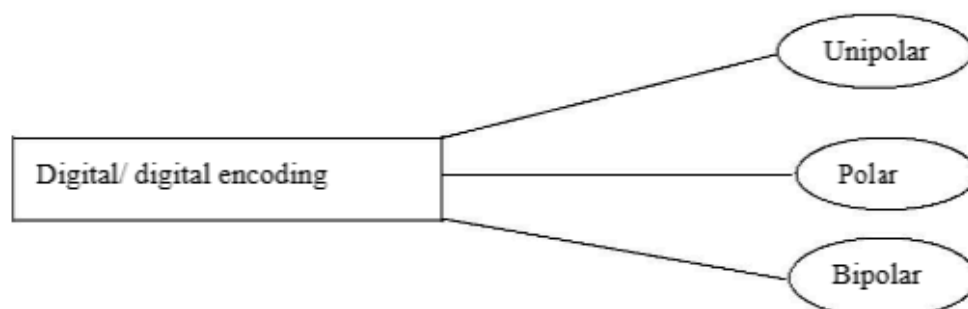
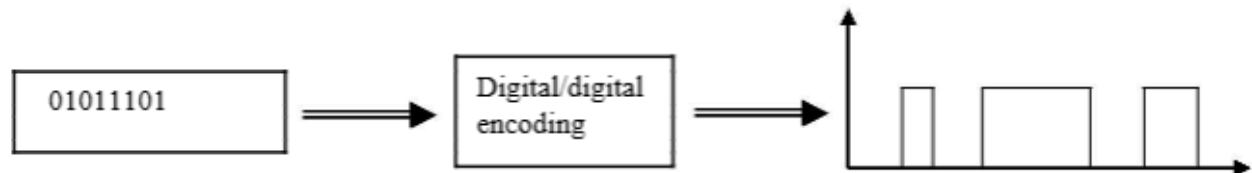
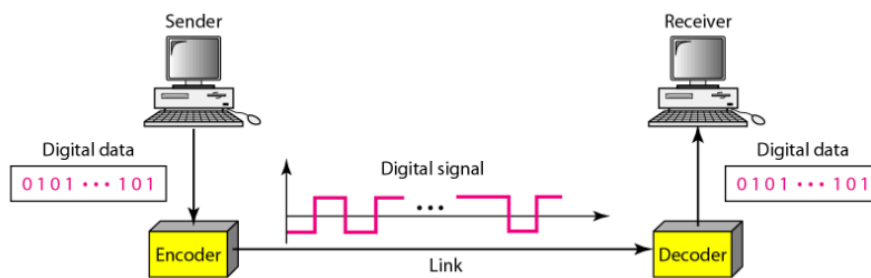
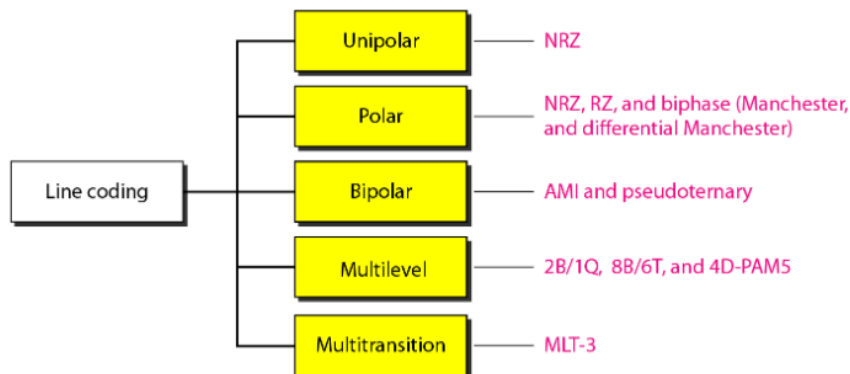


Figure 4.1 *Line coding and decoding*



Although the actual bandwidth of a digital signal is infinite, the effective bandwidth is finite.

Figure 4.4 *Line coding schemes*



6.4.1 Unipolar

Unipolar

Digital transmission systems work by sending voltage pulses along a media link, usually a wire or a cable. In most types of encoding, one voltage level stands for binary 0 and another level stands for binary 1. The polarity of a pulse refers to whether it is positive or negative.

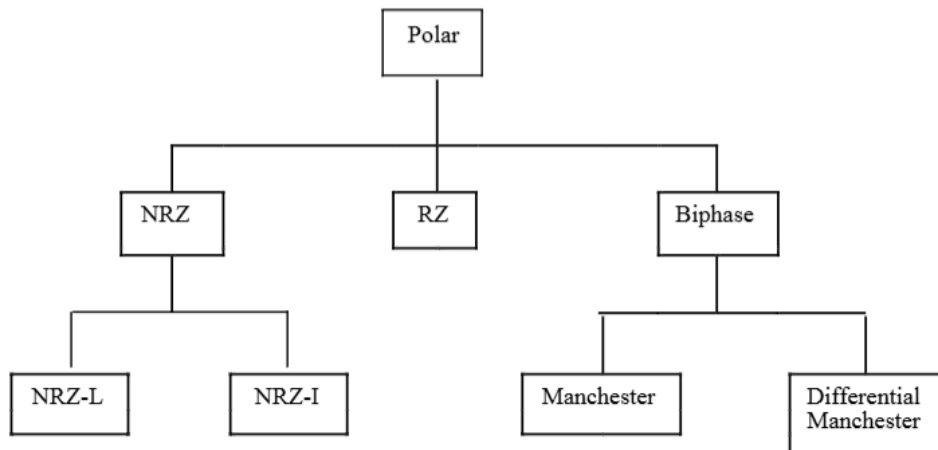
Unipolar encoding is so named because it uses only one polarity. Therefore, only one of the two binary states is encoded, usually the 1. The other state, usually 0, is represented by zero voltage, or an idle line.

Unipolar encoding uses only one level of value.

6.4.2 Polar

Polar Encoding

Polar encoding uses two voltage levels: one positive and one negative. In most polar encoding methods the average voltage level on the line is reduced and the DC component problem of unipolar encoding is alleviated.



6.4.2.1 NRZ : NRZ-I , NRZ-L

Non-Return-to-Zero (NRZ) Encoding

In NRZ encoding, the level of the signal is always either positive or negative. In NRZ system if the line is idle it means no transmission is occurring at all.

- **NRZ-L (Non-return-to-zero, Level)**

In NRZ-L the level of the signal is dependant upon the state of the bit.

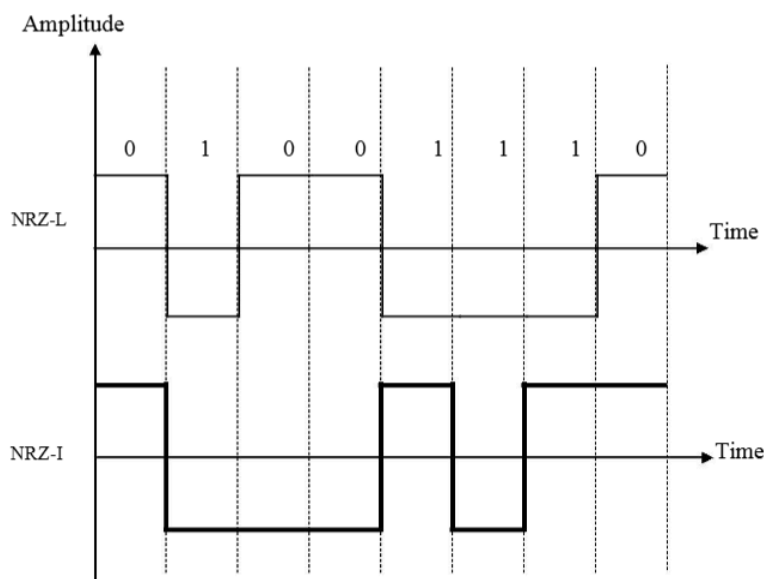
A positive voltage usually means the bit is 0, and negative voltage means the bit is a 1 (and vice versa).

- **NRZ-I (Non-return-to-zero, Invert)**

In NRZ-I, an inversion of the voltage level represents a 1 bit. It is the transition between a positive and a negative voltage, not the voltages themselves that represents a 1 bit. A 0 bit is represented by no change.

An advantage of NRZ-I over NRZ-L is that because the signal changes every time a 1 bit is encountered, it provides some synchronisation.

Each inversion allows the receiver to synchronise its timer to the actual arrival of the transmission.



In NRZ-L the level of the voltage determines the value of the bit.

In NRZ-I the inversion or the lack of inversion determines the value of the bit.

NRZ-L and NRZ-I both have an average signal rate of $N/2$ Bd.

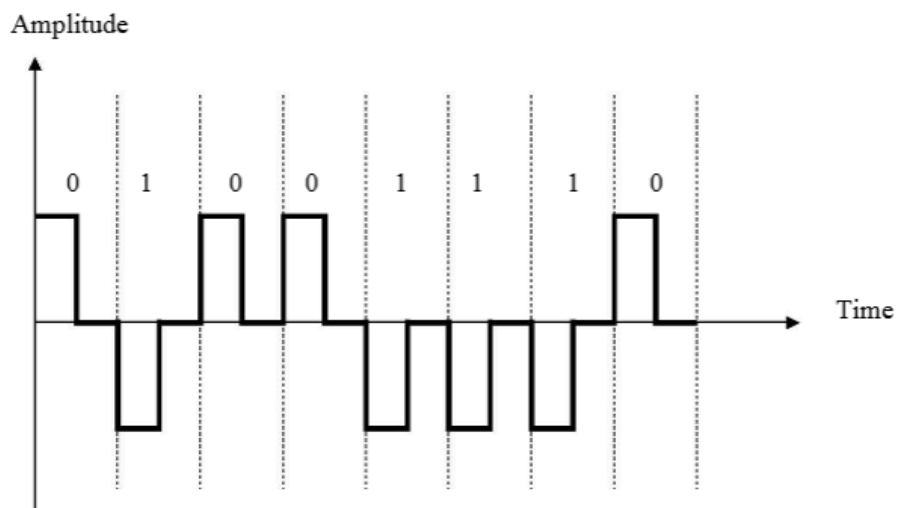
NRZ-L and NRZ-I both have a DC component problem.

6.4.2.2 RZ

RZ (Return-to-zero) Encoding

To assure synchronisation, there must be a signal change for each bit. The receiver can use these changes to built up, update, and synchronise its clock.

One solution is return to zero (RZ) encoding, which uses three Values: positive, negative, and zero.



6.2.2.3 Biphase : Manchester, Differential Manchester,

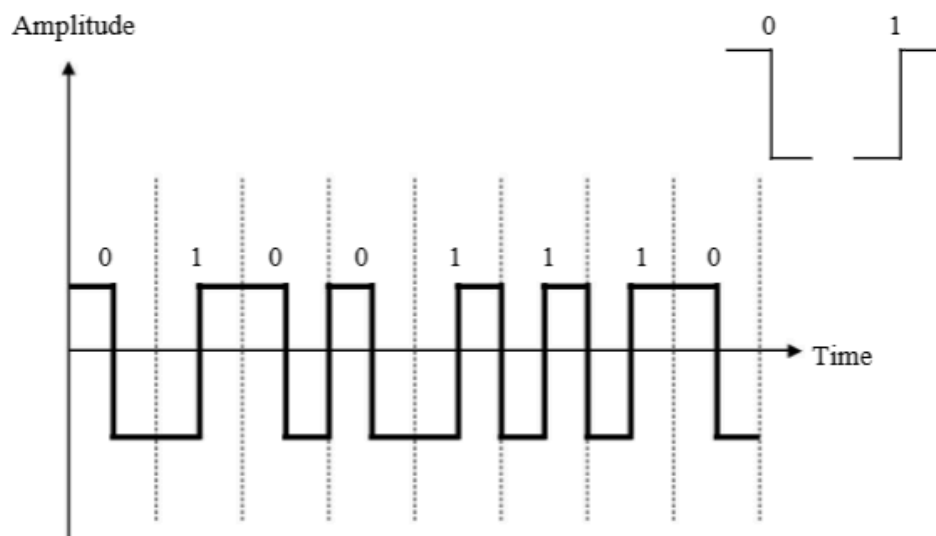
Biphase Encoding

Probably the best existing solution to the problem of synchronisation is biphase encoding. In this method, the signal changes at the middle of the bit interval but does not return to zero. Instead it continues to the opposite pole. As in RZ, these mid-interval transitions allow for synchronisation.

Biphase encoding is implemented in two different ways: Manchester and differential Manchester.

Manchester

Manchester encoding uses the inversion at the middle of each bit interval for both synchronisation and bit representation. A negative-to-positive transition represents binary 1 and a positive-to-negative transition represents binary 0.



Differential Manchester

In this method, the inversion at the middle of the bit is used for synchronisation, but the presence or absence of an additional transition at the beginning of the interval is used to identify the bit. A transition means binary 0 and no transition means binary 1. The bit representation is shown by the inversion and non-inversion at the beginning of the bit.

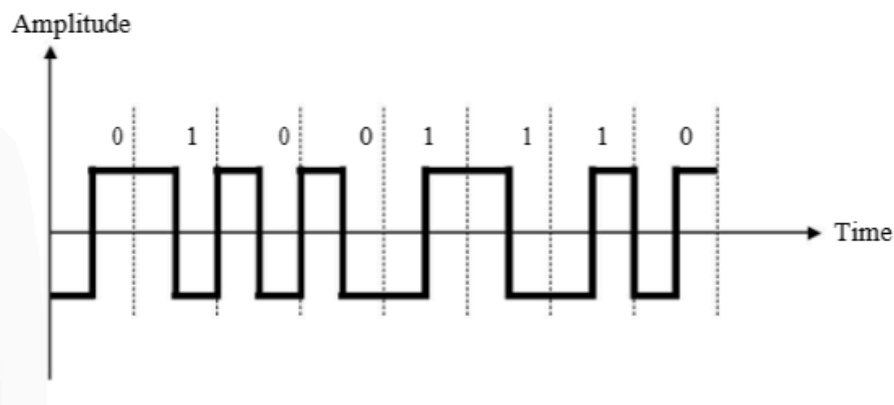
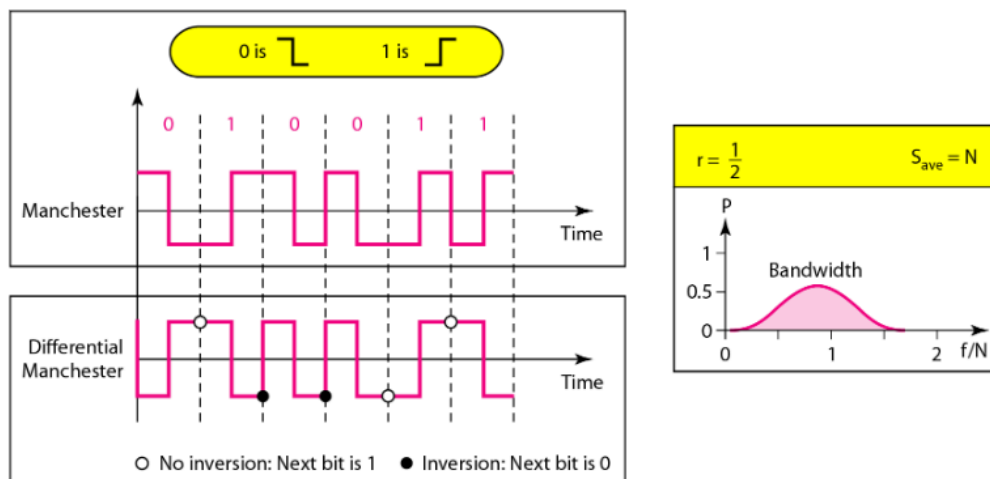


Figure 4.8 Polar biphase: Manchester and differential Manchester schemes



In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.

The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.

6.4.3 Bipolar : B8ZS, HDB3

**In bipolar encoding, we use three levels:
positive, zero, and negative.**

6.4.3.1 B8ZS

Bipolar 8-Zero Substitution (B8ZS)

B8ZS is the convention adopted in North America to provide synchronisation of long strings of 0s. In most situations B8ZS functions identically to bipolar AMI. Bipolar AMI changes poles with every 1 it encounters. These changes provide the synchronisation needed by the receiver, but the signal does not change during a string of 0s, so synchronisation is lost. The solution provided by B8ZS is to force artificial signal changes, called violations

In B8ZS, if eight 0s come one after another, we change the pattern in one of two ways based on the polarity of previous 1.

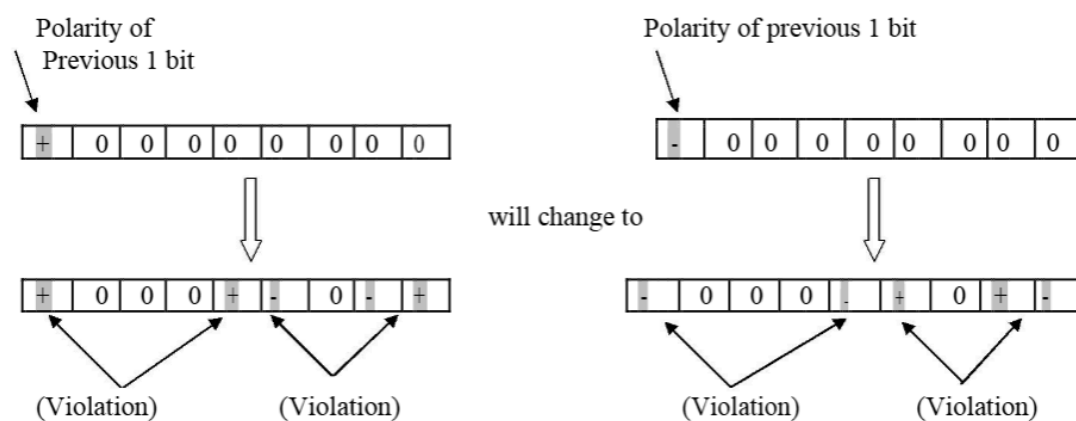
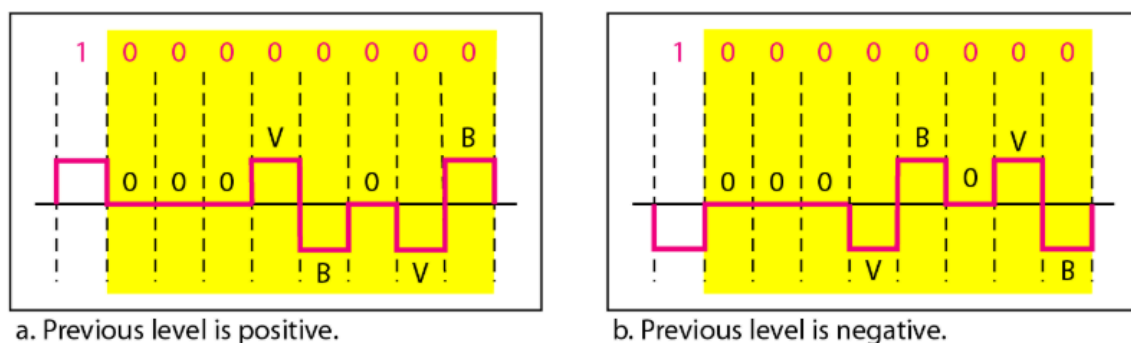


Figure 4.19 Two cases of B8ZS scrambling technique

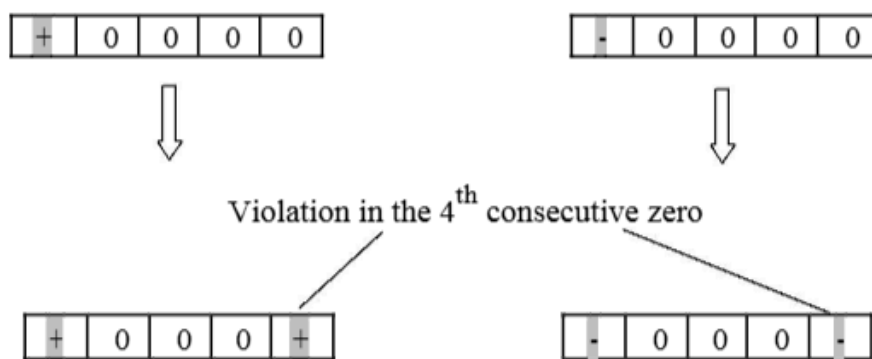


B8ZS substitutes eight consecutive zeros with 000VB0VB.

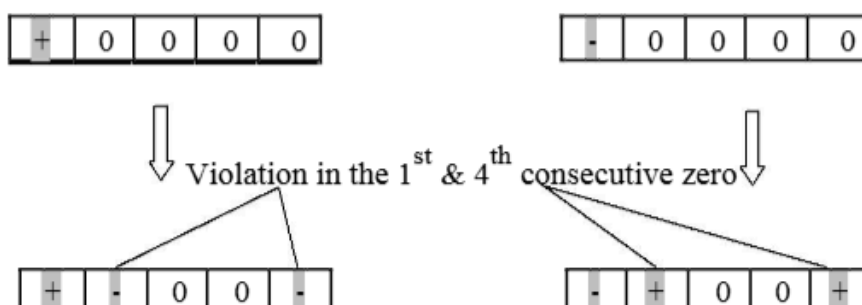
6.4.3.2 HDB3

High-Density Bipolar 3 (HDB3)

In HDB3 if four 0s come one after another, we change the pattern in one of four ways based on the polarity of the previous 1 and the number of 1s since the last substitution.



If the number of 1s since the last substitution is odd



If the number of 1s since the last substitution is even

Using HDB3, encode 10000000000100. Assume that the number of 1s so far is odd and the previous 1 is positive.

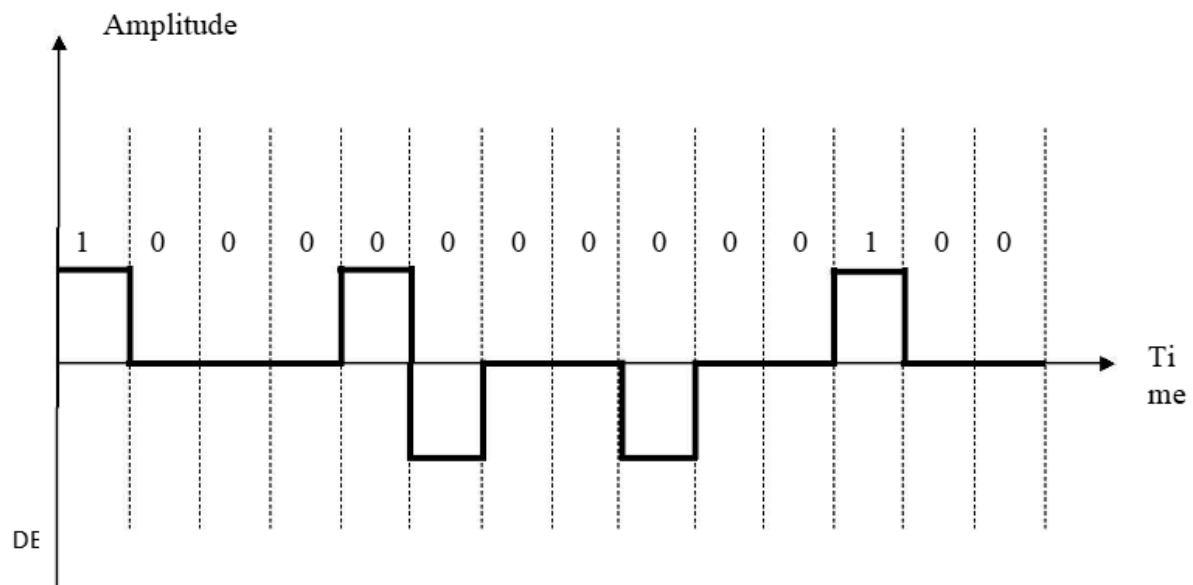
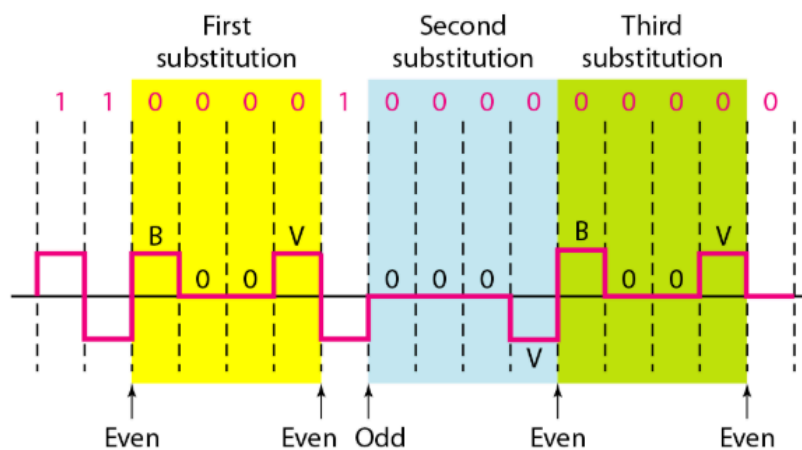


Figure 4.20 *Different situations in HDB3 scrambling technique*



HDB3 substitutes four consecutive zeros with 000V or B00V depending on the number of nonzero pulses after the last substitution.

Table 4.1 *Summary of line coding schemes*

<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s